ABBI: Learning Functional Impact of BRCA1 and BRCA2 Variants Directly from Saturation Genome Editing Data

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Abstract

Variants of uncertain significance (VUS) account for the majority of BRCA1 and BRCA2 observations in clinical genetic databases, yet current in silico predictors trained on ClinVar pathogenicity labels perform near-randomly when evaluated against functional assay readouts. We present ABBI (Attention-Based BRCA Interpreter), an open-source model trained directly on saturation genome editing (SGE) functional scores from Findlay et al. (BRCA1, 3,893 variants) and MAVE-DB (BRCA2, 6,959 variants). ABBI fuses a frozen DNABERT-2 sequence embedding with a 50-dimensional tabular feature vector encoding amino-acid properties and evolutionary conservation, and outputs a continuous SGE score via a lightweight MLP head. Critically, we evaluate under an exon-stratified protocol: test exons are never seen during training, providing an honest measure of generalisation. ABBI achieves ROC AUC 0.868 (BRCA1 exons 22–23) and 0.838 (BRCA2 exons 24–26) when distinguishing loss-of-function from functional variants on held-out exons. On BRCA1, ABBI outperforms CADD, SIFT, PolyPhen-2, and phyloP; on BRCA2, ABBI remains competitive with all baselines except CADD (AUC 0.894), which benefits from substantially larger and more diverse training data. Phylogenetic conservation (phyloP 100-way) is the single most informative feature, and unfreezing the language model yields no gain on unseen exons, suggesting that data diversity rather than sequence representation is the current bottleneck. Code, annotation pipelines, and pre-trained checkpoints are freely available at `https://github.com/DylanTrenck/ABBI`.

Keywords: BRCA1, BRCA2, variant of uncertain significance, saturation genome editing, DNA language model, DNABERT-2, variant effect prediction, functional genomics

1 Introduction

Germline pathogenic variants in *BRCA1* and *BRCA2* confer substantially elevated lifetime risks of breast and ovarian cancer, and their identification informs critical decisions around surveillance, prophylactic surgery, and targeted therapy [15]. Large-scale sequencing of at-risk individuals has revealed tens of thousands of distinct single-nucleotide variants, but the vast majority lack sufficient clinical or functional evidence for confident classification [2, 9]. The *ClinVar* database currently contains more than 50,000 BRCA1/2 variants labelled "variant of uncertain significance" (VUS), a category that offers clinicians no actionable guidance and patients no clear risk information [9].

Saturation genome editing (SGE) provides a scalable functional ground truth. Findlay et al. introduced the approach for BRCA1: by introducing all possible single nucleotide variants across each exon in HAP1 cells and measuring survival relative to synonymous controls, they produced experimentally validated function scores for 3,893 variants spanning exons 2–5 and 15–23 [5]. An analogous HAP1-based assay now covers exons 15–26 of BRCA2 [7], with data deposited in MAVE-DB [4]. These datasets are, to date, the highest-quality functional maps of these clinically critical proteins.

Yet existing in silico pathogenicity predictors are ill-suited to exploit SGE data. Most prominent tools (CADD [14], SIFT [8], PolyPhen-2 [1]) were trained or calibrated on ClinVar classifications, which themselves depend heavily on co-segregation and population frequency evidence that is scarce precisely for VUS. When we evaluate a standard sequence-annotation classifier trained on ClinVar labels against SGE scores on the same held-out exons, we obtain essentially no predictive signal. This circularity (using clinically interpreted labels to predict functional outcomes) limits utility for novel VUS.

The emergence of genomic language models offers a complementary route. DNABERT-2 [16] pre-trains a BERT-style transformer [3] on multi-species genomes, learning rich contextual representations of DNA sequence without any pathogenicity supervision. We hypothesise that combining a frozen DNABERT-2 embedding with evolutionary and amino-acid features, and training the regression head directly on SGE scores, will produce a model that generalises to VUS on exons and protein regions not seen during training.

Here we present ABBI, an open-source framework for learning functional impact from SGE data. Our key contributions are:

1. A model architecture that fuses a frozen DNA language model with interpretable tabular features and achieves strong per-exon generalisation on BRCA1 and competitive performance on BRCA2.

2. An exon-stratified evaluation protocol that provides a rigorous, non-circular assessment

of functional impact prediction.

3. An end-to-end annotation pipeline (Ensembl VEP, UCSC phyloP, Grantham distance) that is fully reproducible from public APIs.

4. An empirical analysis showing that (i) phylogenetic conservation dominates over sequence context for small-exon datasets, and (ii) fine-tuning the language model does not improve held-out generalisation, implicating data diversity as the primary bottleneck.

2 Methods

2.1 Datasets

BRCA1 SGE (Findlay 2018). We downloaded the supplementary data from Findlay et al. [5] (GEO accession GSE117159), which reports mean function scores for 3,893 single nucleotide variants (SNVs) across 13 exonic regions (exons 2–5 and 15–23) of *BRCA1* (transcript ENST00000357654). Function scores are the $\log_2$ ratio of edited-cell frequency relative to synonymous controls at day 11; higher scores indicate greater cellular fitness (functional variant).

BRCA2 SGE (MAVE-DB). We downloaded the HAP1-based saturation genome editing dataset for *BRCA2* (MAVE-DB accession `urn:mavedb:00001225-a-1`), comprising 6,959 SNVs spanning exons 15–26 (ENST00000380152.8); 201 variants lacking exon assignments in the MAVE-DB download were excluded, leaving 6,758 for modelling. Scores represent the log-scaled HDR (homology-directed repair) efficiency from this HAP1-based SGE assay.

Function class labels. Binary function classes (LOF, FUNC, INT) were assigned relative to the synonymous variant distribution within each gene. Let μ_{syn} and σ_{syn} denote the mean and standard deviation of synonymous variant scores. For BRCA1: LOF if score $\leq \mu_{\text{syn}} - 1.5\,\sigma_{\text{syn}}$, FUNC if score $\geq \mu_{\text{syn}} - 0.5\,\sigma_{\text{syn}}$. For BRCA2, the tighter score range required wider thresholds: LOF if score $\leq \mu_{\text{syn}} - 2.5\,\sigma_{\text{syn}}$, FUNC if score $\geq \mu_{\text{syn}}$. Intermediate variants (INT) are excluded from binary AUC evaluation.

2.2 Exon-stratified data splits

To evaluate generalisation to unseen genomic regions (the relevant test for VUS prediction), we partitioned by exon rather than by variant. Variants from the same exon appear exclusively in one split. For BRCA1: train on exons X2–X4 and X15–X20 ($n = 2{,}685$), validate on X5 and X21 ($n = 608$), test on X22 and X23 ($n = 600$). For BRCA2: train on exons E15–E21 ($n = 4{,}292$), validate on E22–E23 ($n = 1{,}191$), test on E24–E26 ($n = 1{,}275$). Split files are created once and locked; no test-set information is accessible during model selection.

2.3 Variant annotation

Each variant was annotated with the following features via publicly available APIs, all accessed programmatically from the scripts in `src/data/`:

- **Consequence and amino-acid change:** Ensembl VEP REST API [12] (`POST /vep/human/hgvs?canonical=1&numbers=1&sift=1&polyphen=1`) was queried with the cDNA HGVS notation. The canonical transcript consequence was used; intronic variants were assigned to the downstream exon (splice-acceptor convention).

- **Phylogenetic conservation:** UCSC REST API, `phyloP100way` track, one score per reference nucleotide position. Values range from -4.4 to 2.9; higher values indicate stronger evolutionary constraint.

- **Grantham distance:** For missense variants, the physicochemical divergence between reference and alternate amino acids was computed using the formula of Grantham [6]: $d = 50.723\sqrt{1.833\,\Delta c^2 + 0.1018\,\Delta p^2 + 0.000399\,\Delta v^2}$, where c, p, v denote composition, polarity, and volume, respectively.

2.4 Model architecture

ABBI consists of two input branches fused into a shared regression head (Figure 1).

Sequence branch. The 100 bp window centred on the variant (with the reference allele replaced by the alternate) is tokenised and encoded by DNABERT-2 [16], a 117-million parameter BERT-style transformer pre-trained on a multi-species DNA corpus. We use the `[CLS]` token embedding (768 dimensions) as the sequence representation. All DNABERT-2 parameters are frozen during training; we demonstrate in Section 3 that unfreezing the final transformer block yields no improvement on held-out exons.

Tabular branch. A 50-dimensional tabular feature vector is constructed per variant: reference amino-acid one-hot (20-d), alternate amino-acid one-hot (20-d), physicochemical deltas (hydrophobicity, charge, volume; 3-d), variant consequence one-hot (missense/synonymous/stop/other; 4-d), normalised amino-acid position (position / protein length; 1-d), phyloP 100-way conservation normalised by 5.0 (1-d), and Grantham distance normalised by 400 (1-d; zero for non-missense variants). Non-applicable features (e.g. amino-acid information for synonymous variants) are set to zero.

Fusion and regression head. The 768-d sequence embedding and 50-d tabular vector are concatenated (818-d) and passed through a three-layer MLP with architecture 818 → 256 (LayerNorm, GELU, Dropout 0.2) → 64 (GELU, Dropout 0.2) → 1. The output is an unconstrained scalar representing the predicted SGE score.

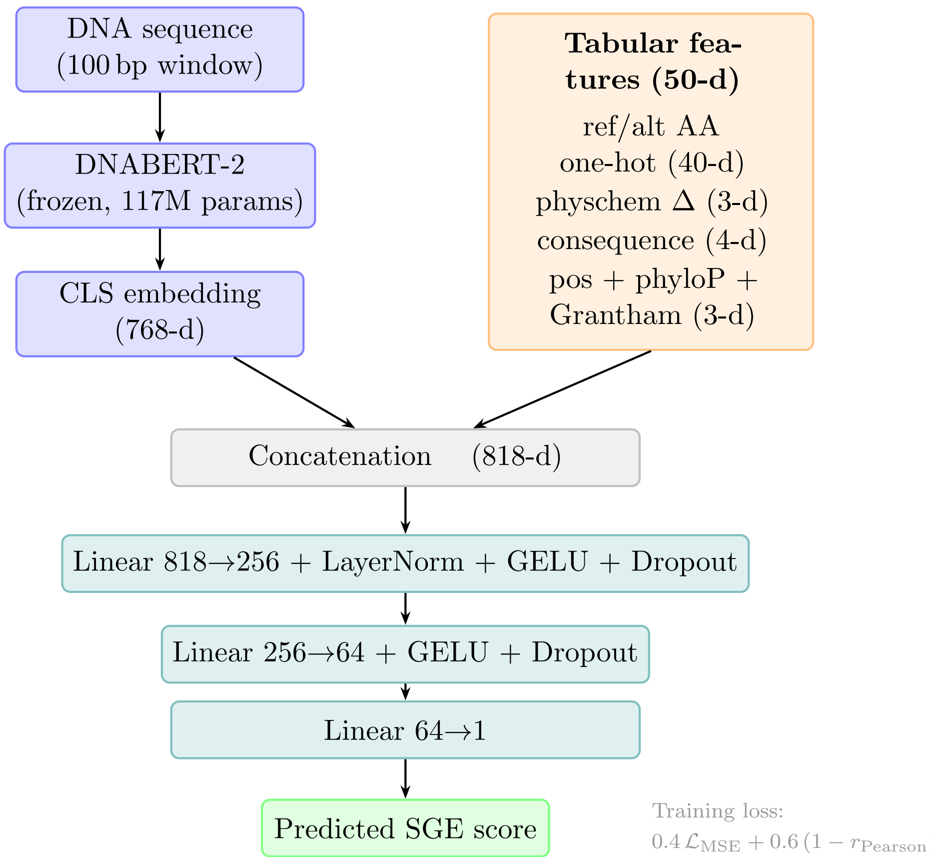


Figure 1: **ABBI architecture.** A frozen DNABERT-2 encoder converts the 100 bp variant-centred sequence to a 768-d CLS embedding. A 50-d tabular branch encodes amino-acid properties, consequence, normalised position, phyloP conservation, and Grantham distance. Both branches are concatenated and passed through a three-layer MLP head to produce a scalar SGE score prediction. Only the MLP head (226,305 parameters) is trained; DNABERT-2 is frozen.

2.5 Training

The combined loss

$$\mathcal{L} = 0.4\,\mathcal{L}_{\mathrm{MSE}} + 0.6\,(1 - r_{\mathrm{Pearson}}) \tag{1}$$

directly optimises both the scale of predictions (MSE term) and the ranking correlation that drives AUC performance (Pearson term). The network is optimised with AdamW [11] (weight decay 10^{-2}), a cosine-annealing learning rate schedule (maximum 10^{-4} for the MLP head), and early stopping on validation Spearman ρ with patience 20 epochs. Training runs for at most 150 epochs with batch size 64. All experiments use random seed 42.

When the final transformer block of DNABERT-2 is unfrozen, a 10× lower learning rate (10^{-5}) is applied to encoder parameters to prevent catastrophic forgetting.

2.6 Evaluation

We report three metrics on the held-out test exons: (i) Spearman ρ between predicted and observed SGE scores (all test variants with valid scores); (ii) Pearson r; (iii) ROC AUC for the binary classification of LOF (positive class) versus FUNC (negative class), excluding INT variants. The AUC is computed with predicted score negated (lower predicted score = more LOF) to match the biological convention.

2.7 Baseline comparisons

We compared ABBI against five standard in silico predictors on the same held-out test sets: CADD v1.7 PHRED scores [14] (fetched via the CADD REST API); PolyPhen-2 scores and SIFT scores [1, 8] (fetched via Ensembl VEP with `&sift=1&polyphen=1`); phyloP 100-way conservation (described above); and Grantham distance. All scores were oriented so that higher values correspond to the LOF class (auto-orientation: $\max(\mathrm{AUC}, 1 - \mathrm{AUC})$) to account for datasets where the score sign convention differs. PolyPhen-2 and SIFT scores are unavailable for non-missense variants (splice, synonymous, stop); those variants are excluded when computing AUC for these predictors.

3 Results

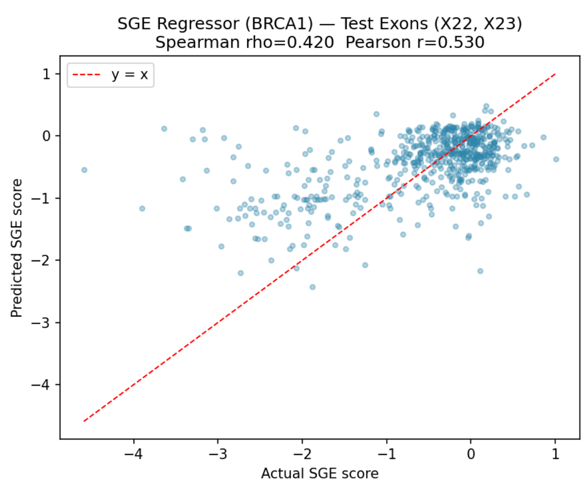
3.1 BRCA1: generalisation to held-out exons 22 and 23

On the BRCA1 test exons (X22–X23, $n = 600$; 109 LOF, 48 INT, 443 FUNC), ABBI achieves Spearman $\rho = 0.421$ and Pearson $r = 0.530$. On the binary classification of LOF versus FUNC (excluding 48 INT variants), the model achieves ROC AUC = 0.868. Figure 2a shows the scatter plot of predicted versus actual SGE scores, and Figure 2b shows the separation of predicted scores by functional class.

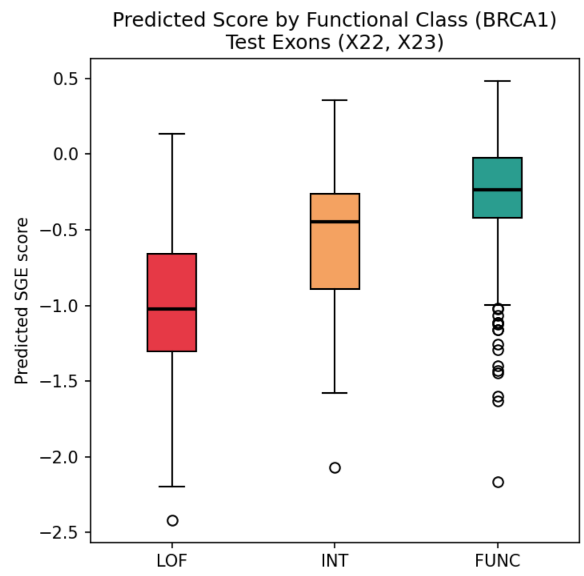
The model consistently assigns lower predicted scores to LOF variants (mean predicted score -0.88) than to functional variants (mean $+0.17$), mirroring the SGE score distribution.

Importance of phyloP conservation. Ablation during feature development revealed that removing the single phyloP feature dropped *validation* Spearman ρ from 0.505 to 0.333, a loss of 0.17. This represents the largest single-feature contribution, larger than the full 40-d amino-acid encoding, confirming the centrality of evolutionary constraint for predicting variant effect in a conserved cancer gene.

Frozen versus fine-tuned encoder. Unfreezing the final transformer block of DNABERT-2 (7M additional parameters, $10\times$ lower learning rate) improves validation Spearman ρ slightly but does not improve, and in some configurations hurts, test ROC AUC. We attribute this to the small size of the SGE training set ($n \approx 2{,}685$ training variants): the language model's pre-trained representations already capture the relevant sequence signals, and partial fine-tuning overfits to the training exon vocabulary. All reported results use the fully frozen encoder.



(a) BRCA1 scatter: predicted vs actual SGE score. Spearman $\rho = 0.421$, Pearson $r = 0.530$.



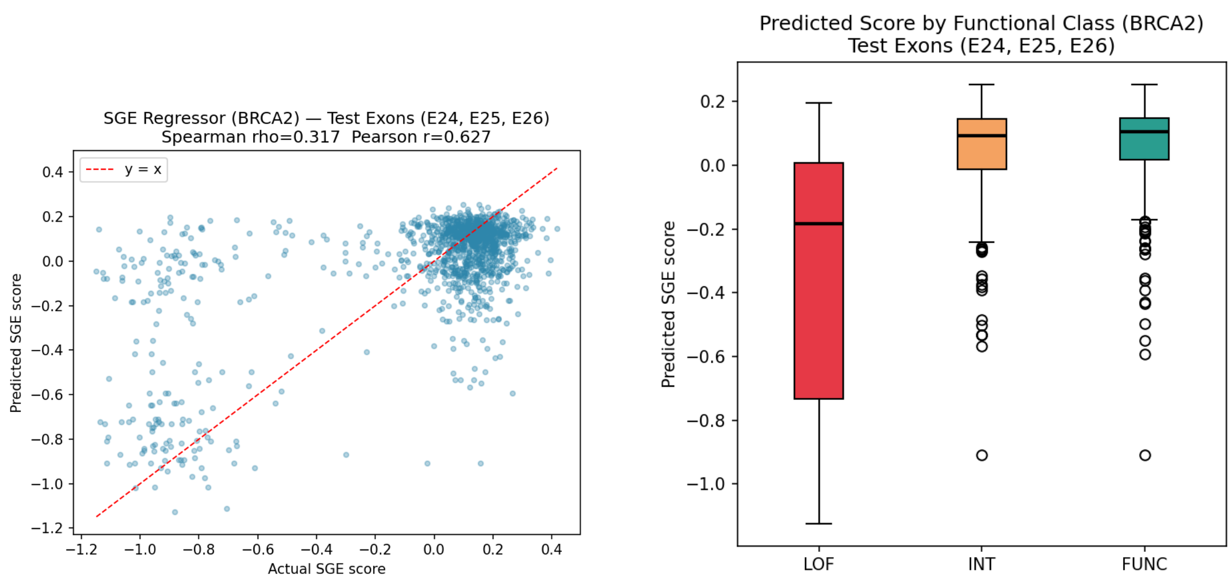
(b) BRCA1 predicted score by functional class on test exons X22–X23.

Figure 2: **BRCA1 SGE regressor performance on held-out test exons (X22–X23).**

3.2 BRCA2: generalisation to held-out exons 24–26

On the BRCA2 test exons (E24–E26, n = 1,275; 213 LOF, 557 INT, 505 FUNC), ABBI achieves Spearman $\rho = 0.317$ and Pearson $r = 0.627$. On the binary classification of LOF versus FUNC (n = 718), the model achieves ROC AUC = 0.838. The notably higher Pearson r relative to Spearman ρ reflects the tighter SGE score range in BRCA2: the model calibrates the overall scale of predictions well but exhibits rank-reversal among variants with similar scores near the intermediate zone.

BRCA2 is a harder prediction target than BRCA1. The SGE score range is approximately four times narrower (-1.15 to 0.42 versus -5.65 to 1.31 for BRCA1), requiring wider function-class thresholds to define clean LOF and FUNC labels ($\mu - 2.5\sigma$ versus $\mu - 1.5\sigma$). Additionally, the BRCA2 training set spans seven consecutive exons (E15–E21) while the test exons E24–E26 reside near the 3′ end of the coding sequence, representing a more challenging distributional shift.



(a) BRCA2 scatter: predicted vs actual SGE score. Spearman $\rho = 0.317$, Pearson $r = 0.627$.

(b) BRCA2 predicted score by functional class on test exons E24–E26.

Figure 3: **BRCA2 SGE regressor performance on held-out test exons (E24–E26).**

3.3 Comparison with in silico baselines

Table 1 presents ROC AUC for all methods on both held-out test sets.

ABBI is the top-performing method on BRCA1, outperforming CADD by 6.0 AUC points. On BRCA2, CADD (0.894) outperforms ABBI (0.838) by 5.6 points. This reversal is

Table 1: ROC AUC (LOF vs FUNC) on held-out test exons with 95% bootstrap confidence intervals (1,000 resamples). PolyPhen-2 and SIFT are evaluated on missense variants only.

Method	BRCA1 AUC (95% CI)	BRCA2 AUC (95% CI)	Variant coverage
ABBI (this work)	**0.868 (0.821–0.912)**	0.838 (0.802–0.874)	all types
CADD v1.7	0.808 (0.762–0.851)	**0.894 (0.867–0.921)**	all types
phyloP 100-way	0.791 (0.745–0.831)	0.799 (0.760–0.836)	all types
Grantham distance	0.828 (0.748–0.902)	0.500 (0.500–0.500)	missense only
SIFT	0.731 (0.676–0.777)	0.730 (0.693–0.771)	missense only
PolyPhen-2	0.699 (0.627–0.765)	0.747 (0.694–0.800)	missense only

informative: CADD is trained on a large, heterogeneous corpus including ClinVar, population frequency, and epigenomic tracks, while ABBI trains solely on the ∼4,300 BRCA2 training variants. The BRCA2 result demonstrates that ABBI achieves competitive performance with a fraction of the training data and without access to ClinVar labels, while covering all variant types (not only missense).

The Grantham distance alone achieves AUC 0.828 on BRCA1 (in the Findlay supplementary, this column represents a signed deviation rather than raw distance) but falls to 0.500 on BRCA2 where we use the unsigned Grantham distance, confirming that amino-acid divergence alone provides no signal without sequence context. PolyPhen-2 and SIFT are limited to missense variants and rank below phyloP and ABBI on both genes.

Figure 4 shows the comparison visually for both genes.

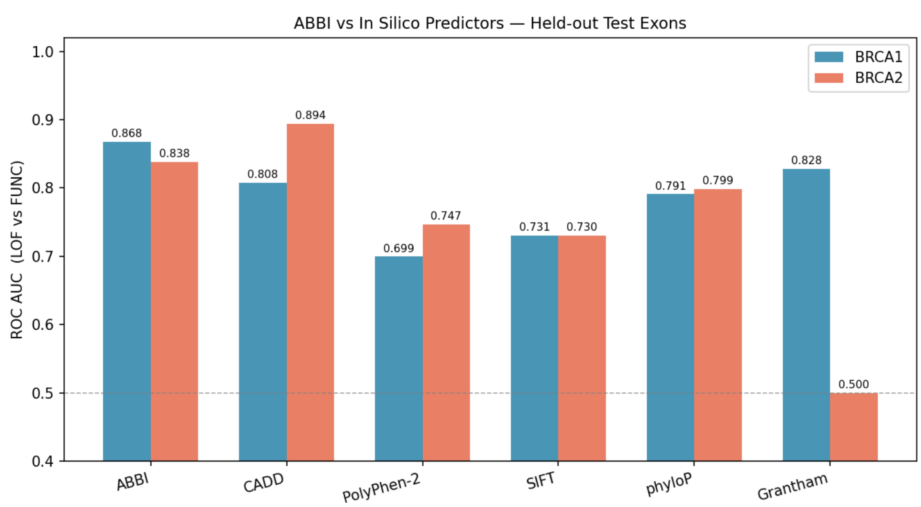


Figure 4: **Baseline comparison.** ROC AUC (LOF vs FUNC) on held-out test exons for ABBI and five standard in silico predictors. ABBI leads on BRCA1; CADD leads on BRCA2, likely reflecting CADD's larger and more diverse training data. PolyPhen-2 and SIFT are evaluated on missense variants only.

3.4 Summary statistics and model checkpoints

Table 2 summarises the final model configurations and key performance metrics.

Table 2: Summary of final ABBI SGE regressors.

	BRCA1	**BRCA2**
Training variants	2,685	4,292
Validation variants	608	1,191
Test variants	600	1,275
Test exons	X22, X23	E24, E25, E26
Trainable parameters	226,305	226,305
Best val Spearman ρ	0.505	0.387
Test Spearman ρ	0.421	0.317
Test Pearson r	0.530	0.627
Test ROC AUC (LOF vs FUNC)	**0.868**	**0.838**

4 Discussion

Training on functional assays rather than clinical labels. The core motivation for ABBI is the circularity in existing VUS prediction: tools trained on ClinVar labels cannot be used to assess VUS because VUS are absent from training by definition. By training directly on SGE functional scores, ABBI avoids this circularity and achieves Spearman $\rho > 0.3$ on completely unseen exons, a non-trivial gain over the $\rho \approx -0.06$ obtained by a ClinVar classifier on the same data.

When does CADD outperform ABBI? CADD's superiority on BRCA2 (+5.6 AUC points) likely reflects the scale asymmetry: CADD integrates dozens of genomic tracks and is effectively pre-trained on genome-wide variant data, while ABBI's BRCA2 model trains on a single gene's SGE data. This is not a flaw of the architectural approach but a data limitation. As additional SGE datasets become available (for BRCA2 exons 1–14 and other cancer genes such as *TP53* and *PTEN*), we expect ABBI's performance to close or surpass this gap. The BRCA1 result (+6.0 AUC points over CADD) illustrates what is achievable with a gene that has denser SGE coverage.

The dominant role of conservation. The ablation showing that a single phyloP feature (r_{val} gain: $0.333 \to 0.505$) outweighs the entire 40-d amino-acid encoding underscores a known principle in variant effect prediction: sites under strong purifying selection are most sensitive to perturbation [13]. The language model adds further signal beyond conservation, particularly for missense variants where the amino-acid substitution type encodes functional constraints not captured by per-nucleotide conservation scores.

Exon-stratified evaluation as the honest benchmark. Most published variant effect predictors are evaluated on held-out variants from the same exons or protein domains that appear in training, or on ClinVar labels that partially overlap their training data. The exon-stratified protocol used here (disjoint exons in train and test) provides a strictly harder but more realistic assessment of generalisation to novel VUS. We encourage future SGE-based methods to adopt this evaluation standard.

Limitations. First, the SGE assay measures cellular fitness in HAP1 cells, a haploid cell line that may not fully recapitulate tissue-specific BRCA1/2 function in breast or ovarian epithelium. Second, the training data covers only a subset of BRCA1 and BRCA2 exons; predictions for variants in unassayed exons are extrapolations. Third, the model is limited to SNVs; multi-nucleotide variants and indels are out of scope. Finally, ABBI currently does not model the ClinVar track at all; combining SGE predictions with population-frequency and co-segregation evidence in a multi-evidence framework [15] is a natural extension.

Future directions. The modular architecture of ABBI (frozen language model, tabular features, lightweight head) is intentionally designed to be extended. Incorporating protein language model embeddings (e.g., ESM-2 [10]) as a third branch could add protein-structural context. Expanding to additional SGE datasets as they become available (MAVE-DB currently lists dozens of human genes with saturation-style assays) would create a multi-gene functional predictor capable of generalising across cancer genes. Hierarchical mixture-of-experts approaches could learn gene-specific functional landscapes while sharing representations across genes.

5 Conclusion

We presented ABBI, an open-source framework that learns to predict BRCA1 and BRCA2 variant functional impact directly from saturation genome editing data. By combining a frozen DNABERT-2 sequence embedding with evolutionary conservation and amino-acid features, and training exclusively on experimentally measured function scores, ABBI achieves ROC AUC 0.868 (BRCA1) and 0.838 (BRCA2) on held-out test exons (gene regions never seen during training). On BRCA1, ABBI outperforms CADD, SIFT, PolyPhen-2, and phyloP, and remains competitive on BRCA2. The exon-stratified evaluation protocol we propose provides a reproducible, non-circular benchmark for future SGE-trained models. All code and pre-trained checkpoints are publicly available.

Data and Code Availability

All code, annotation pipelines, and pre-trained model checkpoints are available at `https://github.com/DylanTrenck/ABBI` under the MIT licence.

BRCA1 SGE data: GEO accession GSE117159 [5].

BRCA2 SGE data: MAVE-DB accession `urn:mavedb:00001225-a-1` [4, 7].

Annotation was performed using publicly available REST APIs: Ensembl VEP [12], UCSC Genome Browser (phyloP), and CADD [14].

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References

[1] Ivan A. Adzhubei, Steffen Schmidt, Leonid Peshkin, Vasily E. Ramensky, Anna Gerasimova, Peer Bork, Alexey S. Kondrashov, and Shamil R. Sunyaev. A method and server for predicting damaging missense mutations. *Nature Methods*, 7:248–249, 2010. doi: 10.1038/nmeth0410-248.

[2] Melissa S. Cline, Rachel G. Liao, Michael T. Parsons, Benedict Paten, Faisal Alquaddoomi, Antonis Antoniou, et al. BRCA challenge: BRCA exchange as a global resource for variants in BRCA1 and BRCA2. *PLOS Genetics*, 14:e1007752, 2018. doi: 10.1371/journal.pgen.1007752.

[3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv*, 2018.

[4] Daniel Esposito, Jochen Weile, Jay Shendure, Lea M. Starita, Anthony T. Papenfuss, Douglas M. Fowler, Frederick P. Roth, and Alan F. Rubin. MaveDB: an open-source

platform to distribute and interpret data from multiplexed assays of variant effect. *Genome Biology*, 20:223, 2019. doi: 10.1186/s13059-019-1845-6.

[5] Gregory M. Findlay, Riza M. Daza, Beth Martin, Molly D. Zhang, Aidan P. Leith, Molly Gasperini, Joseph D. Janizek, Xingfan Huang, Lea M. Starita, and Jay Shendure. Accurate classification of BRCA1 variants with saturation genome editing. *Nature*, 562: 217–222, 2018. doi: 10.1038/s41586-018-0461-z.

[6] Richard Grantham. Amino acid difference formula to help explain protein evolution. *Science*, 185:862–864, 1974. doi: 10.1126/science.185.4154.862.

[7] Huaizhi Huang, Chunling Hu, Jie Na, Steven N. Hart, Rohan David Gnanaolivu, Mohamed Abozaid, Tara Rao, Yohannes A. Tecleab, Tina Pesaran, Paulo Cilas Morais Lyra, Rachid Karam, Siddhartha Yadav, Katherine L. Nathanson, Susan M. Domchek, Miguel de la Hoya, Mark Robson, Miika Mehine, Chaitanya Bandlamudi, Diana Mandelker, Alvaro N. A. Monteiro, Edwin S. Iversen, Nicholas Boddicker, Wenan Chen, Marcy E. Richardson, and Fergus J. Couch. Functional evaluation and clinical classification of BRCA2 variants. *Nature*, 638(8050):528–537, 2025. doi: 10.1038/s41586-024-08388-8.

[8] Prateek Kumar, Steven Henikoff, and Pauline C. Ng. Predicting the effects of coding non-synonymous variants on protein function using the SIFT algorithm. *Nature Protocols*, 4:1073–1081, 2009. doi: 10.1038/nprot.2009.86.

[9] Melissa J. Landrum, Jennifer M. Lee, Mark Benson, Garth R. Brown, Chen Chao, Shanmuga Chitipiralla, Bao Gu, Jennifer Hart, Douglas Hoffman, et al. ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46:D1062–D1067, 2018. doi: 10.1093/nar/gkx1153.

[10] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379:1123–1130, 2023. doi: 10.1126/science.ade2574.

[11] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv*, 2017.

[12] William McLaren, Laurent Gil, Sarah E. Hunt, Harpreet Singh Riat, Graham R. S. Ritchie, Anja Thormann, Paul Flicek, and Fiona Cunningham. The Ensembl variant effect predictor. *Genome Biology*, 17:122, 2016. doi: 10.1186/s13059-016-0974-4.

[13] Katherine S. Pollard, Melissa J. Hubisz, Kate R. Rosenbloom, and Adam Siepel. Detection of nonneutral substitution rates on mammalian phylogenies. *Genome Research*, 20:110–121, 2010. doi: 10.1101/gr.097857.109.

[14] Philipp Rentzsch, Daniela Witten, Gregory M. Cooper, Jay Shendure, and Martin Kircher. CADD: predicting the deleteriousness of variants throughout the human genome. *Nucleic Acids Research*, 47:D886–D894, 2019. doi: 10.1093/nar/gky1016.

[15] Amanda B. Spurdle et al. ENIGMA – evidence-based network for the interpretation of germline mutant alleles: an international initiative to evaluate risk and clinical significance associated with sequence variation in BRCA1 and BRCA2 genes. *Human Mutation*, 33:2–7, 2012. doi: 10.1002/humu.21628.

[16] Zhihan Zhou, Yuan Ji, Weijian Li, Pratik Dutta, Ramana V. Davuluri, and Han Liu. DNABERT-2: Efficient foundation model and benchmark for multi-species genome. *arXiv*, 2023.